

Largest Concatenation-Component in Regular Expressions

Hyperscan implements algorithms to decompose regexes into literals and FA in its Rose engine. There may be a relation between how well a regex can be decomposed and the size of its largest concatenation-component.

1.1 Recurrences

Let $\mathcal{A}$ be a finite alphabet and $\mathcal{T}$ be the set of all regexes excluding ϵ over $\mathcal{A}$. The size function $|\cdot|$ is the number of nodes.

Definition 1 (BST-like distribution for regexes). *We define $p : \mathcal{T} \rightarrow [0, 1]$ as follows.*

1. $\sum_{c \in \mathcal{A}} p(c) = 1,$
2. $p \left(\begin{array}{c} \bullet \\ \swarrow \searrow \\ T_1 \quad T_2 \end{array} \right) = \frac{u}{n-2} p(T_1) p(T_2), \text{ where } |T_1| + |T_2| = n - 1.$
3. $p \left(\begin{array}{c} + \\ \swarrow \searrow \\ T_1 \quad T_2 \end{array} \right) = \frac{v}{n-2} p(T_1) p(T_2), \text{ where } |T_1| + |T_2| = n - 1.$
4. $p \left(\begin{array}{c} * \\ | \\ T \end{array} \right) = (1 - u - v) p(T).$

We want to compute the expected size of the largest all-concatenation component in a regex over all of equivalent forms of its parse tree. The reason is as follows. Take the regex

$$(abc + d)(xy + z) = abcxy + abcxz + dxy + dz.$$

We expect the largest concatenation-component to be in $abcxy$ instead of abc because the former is extracted by Hyperscan. Therefore, a subtree may *yield* its concatenation-component to its parent if the parent is a concatenation node.

Recall the construction

$$\mathcal{T} = \sum_{c \in \mathcal{A}} c + \begin{array}{c} * \\ | \\ \mathcal{T} \end{array} + \begin{array}{c} \bullet \\ \swarrow \searrow \\ \mathcal{T} \quad \mathcal{T} \end{array} + \begin{array}{c} + \\ \swarrow \searrow \\ \mathcal{T} \quad \mathcal{T} \end{array}.$$

Define the random variables

1. M_n : the largest concatenation-component that is not inside a Kleene star of a regex of size n *itself*,
2. Y_n : the largest concatenation-component that is not inside a Kleene star of a regex of size n that it *yields* to its parent.

We have the following recurrences

$$Y_{n+2} = \begin{cases} Y_{U_n}^{(1)} + Y_{n+1-U_n}^{(2)} + 1 & \text{prob. } u, \\ \max(Y_{U_n}^{(1)}, Y_{n+1-U_n}^{(2)}) & \text{prob. } v, \\ 0 & \text{prob. } 1 - u - v \end{cases} \quad Y_1 = Y_2 = 0. \quad (1.1)$$

$$M_{n+2} = \begin{cases} \max(Y_{U_n}^{(1)} + Y_{n+1-U_n}^{(2)} + 1, M_{U_n}^{(1)}, M_{n+1-U_n}^{(2)}) & \text{prob. } u, \\ \max(M_{U_n}^{(1)}, M_{n+1-U_n}^{(2)}) & \text{prob. } v \\ 0 & \text{prob. } 1 - u - v \end{cases} \quad M_1 = M_2 = 0. \quad (1.2)$$

We have used $Y_n^{(1)}, Y_n^{(2)}, M_n^{(1)}, M_n^{(2)}$ as independent copies of Y_n and M_n respectively, and U_n as a uniform random variable over $\{1, 2, \dots, n\}$ independent of all $Y_n^{(i)}$ and $M_n^{(i)}$.

1.2 The yield recurrence

1.2.1 Lower and upper bounds

We rewrite the recurrence (1.1) as

$$Y_{n+2} \stackrel{d}{=} I \left(Y_{U_n}^{(1)} + Y_{n+1-U_n}^{(2)} + 1 \right) + J \max \left(Y_{U_n}^{(1)}, Y_{n+1-U_n}^{(2)} \right), \quad (1.3)$$

where I and J are independent Bernoulli random variables taking value 1 with probability u and v respectively such that $I + J \leq 1$, and $Y_n^{(1)}, Y_n^{(2)}$ are independent copies of Y_n . The notion of equality is in distribution. Note that we can reuse the same $Y_n^{(1)}, Y_n^{(2)}$ and U_n for the second term because the two terms are mutually exclusive.

Let $y_n = \mathbb{E}[Y_n]$. Take expectation on both sides, we have

$$y_{n+2} = u + \frac{2u}{n} \sum_{i=1}^n y_i + \frac{v}{n} \sum_{i=1}^n \mathbb{E}[\max(Y_i, Y_{n+1-i})].$$

For an upper bound, we use the fact that $\max(Y_i, Y_{n+1-i}) \leq Y_i + Y_{n+1-i}$, which gives an upper bound sequence

$$y_{n+2}^U = u + \frac{2(u+v)}{n} \sum_{i=1}^n y_i^U, \quad y_1^U = y_2^U = 0. \quad (1.4)$$

For a lower bound, we use

$$\begin{aligned} \mathbb{E}[\max(Y_i, Y_{n+1-i})] &= \mathbb{E} \left[\frac{1}{2} (Y_i + Y_{n+1-i} + |Y_i - Y_{n+1-i}|) \right] \\ &\geq \frac{1}{2} (\mathbb{E}[Y_i] + \mathbb{E}[Y_{n+1-i}]) + \frac{1}{2} |\mathbb{E}[Y_i] - \mathbb{E}[Y_{n+1-i}]| \\ &= \frac{1}{2} (y_i + y_{n+1-i} + |y_i - y_{n+1-i}|). \end{aligned}$$

This gives the lower bound sequence

$$y_{n+2}^L = u + \frac{2u+v}{n} \sum_{i=1}^n y_i^L + \frac{v}{2n} \sum_{i=1}^n |y_i^L - y_{n+1-i}^L|, \quad y_1^L = y_2^L = 0. \quad (1.5)$$

Remark. Another lower bound is obtained as follows. By Jensen's inequality

$$\mathbb{E}[\max(Y_i, Y_{n+1-i})] \geq \max(\mathbb{E}[Y_i], \mathbb{E}[Y_{n+1-i}]) = \max(y_i, y_{n+1-i}) = y_{\max(i, n+1-i)},$$

where the last equality holds because y_n is non-decreasing in n . This gives the following lower bound sequence

$$y_{n+2}^L = u + \frac{2u}{n} \sum_{i=1}^n y_i^L + \frac{2v}{n} \sum_{i=\lfloor (n+1)/2 \rfloor}^n y_i^L, \quad y_1^L = y_2^L = 0. \quad (1.6)$$

Looking at the bounds (1.5), we know that when $2u + v > 1$, (y_n) grows at least polynomially.

1.2.2 Asymptotic when both bounds are at least polynomial

Consider the case $2u + v > 1$. By the bounds, there exists

$$\alpha = \inf \left\{ \beta : \frac{Y_n}{n^\beta} \xrightarrow{d} 0 \right\}.$$

We know that $\alpha \in (2u + v - 1, 2(u + v) - 1]$. We want to see if indeed $y_n \sim Cn^\alpha$ for some $C > 0$.

We try the contraction method [2, 3, 1]. Let us mention some conventions. For a random variable X , denote by $\mathcal{L}(X)$ its distribution. Let $(\mathcal{M}, d)$ be a metric space where $\mathcal{M}$ is a set of probability distributions of our random variables and the metric d is such that $d(\mathcal{L}(X), \mathcal{L}(Y)) = 0$ implies $X \stackrel{d}{=} Y$. The idea is as follows

1. Guess a normalization X_n of Y_n that converges in distribution.
2. Plug X_n into the recurrence. Assume that X_n converges, show that the limit equation has a unique solution X^* . This is done by proving that the operator T defined by the right-hand side of the limit equation is a contraction.
3. Prove that indeed $d(\mathcal{L}(X_n), \mathcal{L}(X^*)) \rightarrow 0$.

The method is use to obtain more information about the limit distribution of X_n rather than just its expectation. For example, the following is the recurrence for the number of comparisons in Quicksort where the first element is chosen as the pivot [1]

$$Y_n = Y_{U_n}^{(1)} + Y_{n-1-U_n}^{(2)} + n - 1.$$

One can find the asymptotic of $\mathbb{E}[Y_n]$ by taking expectation on both sides. The contraction is then used to show that $X_n = \frac{Y_n - \mathbb{E}[Y_n]}{n} \rightarrow X^*$, where the denominator n is the “guessed” variance.

Our case is weaker. The normalization is $X_n = Y_n/n^\alpha$ such that $X_n \xrightarrow{d} X^*$. Plugging into the recurrence (1.3), we have

$$\begin{aligned} (n+2)^\alpha X_{n+2} &\stackrel{d}{=} I \left(U_n^\alpha X_{U_n}^{(1)} + (n+1-U_n)^\alpha X_{n+1-U_n}^{(2)} + 1 \right) \\ &\quad + J \max \left(U_n^\alpha X_{U_n}^{(1)}, (n+1-U_n)^\alpha X_{n+1-U_n}^{(2)} \right) \end{aligned}$$

Equivalently,

$$\begin{aligned} X_{n+2} &\stackrel{d}{=} I \left(\left(\frac{U_n}{n+2} \right)^\alpha X_{U_n}^{(1)} + \left(\frac{n+1-U_n}{n+2} \right)^\alpha X_{n+1-U_n}^{(2)} + \frac{1}{(n+2)^\alpha} \right) \\ &\quad + J \max \left(\left(\frac{U_n}{n+2} \right)^\alpha X_{U_n}^{(1)}, \left(\frac{n+1-U_n}{n+2} \right)^\alpha X_{n+1-U_n}^{(2)} \right) \end{aligned} \quad (1.7)$$

Proposition 1. *The right-hand side of the above recurrence converges in distribution to*

$$I \left(U^\alpha X^{(1)} + (1 - U)^\alpha X^{(2)} \right) + J \max \left(U^\alpha X^{(1)}, (1 - U)^\alpha X^{(2)} \right) \quad (1.8)$$

Lemma 1. *Let U_n be a uniform random variable on $\{1, 2, \dots, n\}$ and let U be a uniform random variable on $[0, 1]$. Then $\frac{U_n}{n} \xrightarrow{d} U$.*

Proof. Recall that the CFD of U is

$$F_U(x) = \begin{cases} 0 & x < 0, \\ x & 0 \leq x \leq 1, \\ 1 & x > 1. \end{cases}$$

Let $V_n = U_n/n$. We have $F_{V_n}(x) = \mathbb{P}(U_n/n \leq x) = \mathbb{P}(U_n \leq nx)$. Hence

- If $x \leq 0$, then $F_{V_n}(x) = 0 = F_U(x)$, since $U_n \geq 1$.
- If $x > 1$, then $F_{V_n}(x) = 1 = F_U(x)$, since $U_n \leq n$.
- If $0 < x \leq 1$, then $F_{V_n}(x) = \frac{\lfloor nx \rfloor}{n}$. Since $x - \frac{1}{n} \leq \frac{\lfloor nx \rfloor}{n} \leq x$, we have $F_{V_n}(x) \rightarrow x = F_U(x)$.

□

Back to our case. By Slutsky's theorem, we have $\frac{U_n}{n+2} = \frac{U_n}{n} \cdot \frac{n}{n+2} \xrightarrow{d} U$ as $n \rightarrow \infty$. Similarly, we have $\frac{n+1-U_n}{n+2} \xrightarrow{d} 1-U$.

Lemma 2. *Let (X_n) be a sequence of random variables converging in distribution to X . Let (N_n) be a integer-valued sequence of random variables independent of (X_n) such that $N_n \xrightarrow{d} \infty$. Then $X_{N_n} \xrightarrow{d} X$.*

Proof. We want to show that $\lim_{n \rightarrow \infty} \mathbb{P}(X_{N_n} \leq x) = \mathbb{P}(X \leq x)$ for all x at which F_X is continuous. The CDF of X_{N_n} is

$$\mathbb{P}(X_{N_n} \leq x) = \sum_{k=1}^{\infty} \mathbb{P}(X_k \leq x, N_n = k) = \sum_{k=1}^{\infty} \mathbb{P}(N_n = k) \mathbb{P}(X_k \leq x),$$

where the last equality holds because X_k and N_n are independent. Let $\epsilon > 0$. Since $X_n \xrightarrow{d} X$, there exists K such that for all $k \geq K$, we have

$$|\mathbb{P}(X_k \leq x) - \mathbb{P}(X \leq x)| < \frac{\epsilon}{2}.$$

Since $\sum_{k=1}^{\infty} \mathbb{P}(N_n = k) = 1$, we have

$$\begin{aligned}
|\mathbb{P}(X_{N_n} \leq x) - \mathbb{P}(X \leq x)| &= \left| \sum_{k=1}^{\infty} \mathbb{P}(N_n = k) (\mathbb{P}(X_k \leq k) - \mathbb{P}(X \leq x)) \right| \\
&\leq \sum_{k=1}^{\infty} \mathbb{P}(N_n = k) |\mathbb{P}(X_k \leq k) - \mathbb{P}(X \leq x)| \\
&\leq \sum_{k=1}^{K-1} \mathbb{P}(N_n = k) |\mathbb{P}(X_k \leq k) - \mathbb{P}(X \leq x)| \\
&\quad + \sum_{k=K}^{\infty} \mathbb{P}(N_n = k) |\mathbb{P}(X_k \leq k) - \mathbb{P}(X \leq x)| \\
&\leq \sum_{k=1}^{K-1} \mathbb{P}(N_n = k) \cdot 1 + \sum_{k=K}^{\infty} \mathbb{P}(N_n = k) \cdot \frac{\epsilon}{2}
\end{aligned}$$

Since $N_n \xrightarrow{d} \infty$, we have $\mathbb{P}(N_n \geq K) \rightarrow 1$ as $n \rightarrow \infty$. Hence, there exists N such that for all $n \geq N$, we have $\mathbb{P}(N_n \geq K) > 1 - \frac{\epsilon}{2}$. Therefore, for all $n \geq N$, we have

$$|\mathbb{P}(X_{N_n} \leq x) - \mathbb{P}(X \leq x)| < \frac{\epsilon}{2} + \frac{\epsilon}{2} \left(1 - \frac{\epsilon}{2}\right) \leq \epsilon.$$

Since ϵ is arbitrary, the result follows. $\square$

We want to show that

$$\left(\frac{U_n}{n+2}\right)^{\alpha} X_{U_n}^{(1)} + U^{\alpha} X^{(1)} + \left(\frac{n+1-U_n}{n+2}\right)^{\alpha} X_{n+1-U_n}^{(2)} \xrightarrow{d} U^{\alpha} X^{(1)} + (1-U)^{\alpha} X^{(2)}.$$

We already have the function $(u, v, x, y) \mapsto u^{\alpha}x + v^{\alpha}y$ being continuous. In order to use the Continuous Mapping Theorem, we need to show that the joint distribution of

$$\left(\frac{U_n}{n+2}, \frac{n+1-U_n}{n+2}, \frac{X_{U_n}^{(1)}}{n+2}, \frac{X_{n+1-U_n}^{(2)}}{n+2}\right)$$

converges.

Proposition 2. *Let $T : \mathcal{M}_1 \rightarrow \mathcal{M}_1$ be an operator in the space of probability distributions with finite first moment, defined by*

$$T(\mu) = \mathcal{L} \left(I(U^{\alpha} X^{(1)} + (1-U)^{\alpha} X^{(2)}) + J \max(U^{\alpha} X^{(1)}, (1-U)^{\alpha} X^{(2)}) \right),$$

where $X^{(1)}, X^{(2)} \stackrel{iid}{\sim} \mu$ and U, I, J are independent of $X^{(1)}, X^{(2)}$, defined as in (1.3).

Then T is a contraction in the $(\mathcal{M}_p, \ell_p)$ space with $p \geq 2$, where the ℓ_p metric is defined by

$$\ell_p(\mu, \nu) = \inf \{ \mathbb{E}[|X - Y|^p] : X \sim \mu, Y \sim \nu \}.$$

Proof. Let $\mu, \nu \in \mathcal{M}_1$. Let $X, X^{(1)}, X^{(2)} \stackrel{\text{iid}}{\sim} \mu$ and $Y, Y^{(1)}, Y^{(2)} \stackrel{\text{iid}}{\sim} \nu$ be independent of each other. We have

$$\begin{aligned} \ell_1(T(\mu), T(\nu)) &\leq \mathbb{E} \left[\left| I(U^\alpha X^{(1)} + (1-U)^\alpha X^{(2)}) + J \max(U^\alpha X^{(1)}, (1-U)^\alpha X^{(2)}) \right. \right. \\ &\quad \left. \left. - I(U^\alpha Y^{(1)} + (1-U)^\alpha Y^{(2)}) - J \max(U^\alpha Y^{(1)}, (1-U)^\alpha Y^{(2)}) \right| \right] \\ &\leq \mathbb{E} \left[\left| I \left(U^\alpha X^{(1)} + (1-U)^\alpha X^{(2)} - U^\alpha Y^{(1)} - (1-U)^\alpha Y^{(2)} \right) \right| \right] \\ &\quad + \mathbb{E} \left[\left| J \left(\max(U^\alpha X^{(1)}, (1-U)^\alpha X^{(2)}) - \max(U^\alpha Y^{(1)}, (1-U)^\alpha Y^{(2)}) \right) \right| \right] \end{aligned}$$

Consider the first term. We have

$$\begin{aligned} &\mathbb{E} \left[\left| I \left(U^\alpha X^{(1)} + (1-U)^\alpha X^{(2)} - U^\alpha Y^{(1)} - (1-U)^\alpha Y^{(2)} \right) \right| \right] \\ &= \mathbb{E} \left[I \left| U^\alpha X^{(1)} + (1-U)^\alpha X^{(2)} - U^\alpha Y^{(1)} - (1-U)^\alpha Y^{(2)} \right| \right] \quad (I \geq 0) \\ &= u \mathbb{E} \left[\left| U^\alpha (X^{(1)} - Y^{(1)}) + (1-U)^\alpha (X^{(2)} - Y^{(2)}) \right| \right] \quad (\text{independence of } I) \\ &\leq u \left(\mathbb{E} \left[U^\alpha \left| X^{(1)} - Y^{(1)} \right| \right] + \mathbb{E} \left[(1-U)^\alpha \left| X^{(2)} - Y^{(2)} \right| \right] \right) \quad (U, 1-U \geq 0) \\ &= \frac{2u}{\alpha+1} \mathbb{E}[|X - Y|]. \end{aligned}$$

We turn to the second term. Recall that $|\max(a, b) - \max(c, d)| \leq \max(|a - c|, |b - d|)$ and $\max(a, b) = \frac{1}{2}(a + b + |a - b|)$ for any real numbers a, b, c, d . We have

$$\begin{aligned} &\mathbb{E} \left[\left| J \left(\max(U^\alpha X^{(1)}, (1-U)^\alpha X^{(2)}) - \max(U^\alpha Y^{(1)}, (1-U)^\alpha Y^{(2)}) \right) \right| \right] \\ &\leq v \mathbb{E} \left[\max \left(\left| U^\alpha (X^{(1)} - Y^{(1)}) \right|, \left| (1-U)^\alpha (X^{(2)} - Y^{(2)}) \right| \right) \right] \\ &= \frac{v}{2} \left(\mathbb{E} [U^\alpha] \mathbb{E} \left[\left| X^{(1)} - Y^{(1)} \right| \right] + \mathbb{E} [(1-U)^\alpha] \mathbb{E} \left[\left| X^{(2)} - Y^{(2)} \right| \right] \right. \\ &\quad \left. + \mathbb{E} \left[\left| U^\alpha (X^{(1)} - Y^{(1)}) - (1-U)^\alpha (X^{(2)} - Y^{(2)}) \right| \right] \right) \\ &= \frac{v}{\alpha+1} \mathbb{E}[|X - Y|] + \frac{v}{2} \mathbb{E} \left[\left| U^\alpha (X^{(1)} - Y^{(1)}) - (1-U)^\alpha (X^{(2)} - Y^{(2)}) \right| \right] \end{aligned}$$

Combining the bounds for both terms, we get

$$\ell_1(T(\mu), T(\nu)) \leq \frac{2u}{\alpha+1} \mathbb{E}[|X - Y|] + J \mathbb{E}[|X - Y|].$$

Since $\frac{2u}{\alpha+1} + J < 1$, the operator T is a contraction in the $(\mathcal{M}_1, \ell_1)$ space.

□

Asymptotic Analysis of Some Sequences

In this section, we analyze the asymptotic behavior of some recurrences that arise in the analysis of regular expressions.

Proposition 3. *Consider the recurrence*

$$x_{n+2} = \frac{a}{n} \sum_{i=1}^n x_i + b, \quad n \geq 1, \quad x_1 = x_2 = 0. \quad (2.1)$$

- If $a < 1$, then $x_n \sim \frac{b}{1-a}$.
- If $a = 1$, then $x_n \sim b \ln n$.
- If $a > 1$, then $x_n \sim \frac{b}{(a-1)\Gamma(a)} n^{a-1}$.

Proof. Consider the generating function $X(z) = \sum_{n=1}^{\infty} x_n z^n$ with the convention that $x_0 = 0$. We rewrite the recurrence as

$$(n+2)x_{n+2} - 2x_{n+2} = a \sum_{i=1}^n x_i + bn.$$

We multiply both sides by z^{n+2} and sum over $n \geq 1$. The left-hand side becomes

$$\begin{aligned} \sum_{n=1}^{\infty} ((n+2)x_{n+2} - 2x_{n+2})z^{n+2} &= \sum_{n=3}^{\infty} nx_n z^n - 2 \sum_{n=3}^{\infty} x_n z^n \\ &= z \sum_{n=1}^{\infty} nx_n z^{n-1} - 2 \sum_{n=1}^{\infty} x_n z^n \\ &= zX'(z) - 2X(z). \end{aligned}$$

The right-hand side becomes

$$\begin{aligned} \sum_{n=1}^{\infty} (a \sum_{i=1}^n x_i + bn)z^{n+2} &= az^2 \sum_{n=1}^{\infty} \sum_{i=1}^n x_i z^n + bz^2 \sum_{n=1}^{\infty} nz^n \\ &= az^2 \frac{X(z)}{1-z} + \frac{bz^3}{(1-z)^2}. \end{aligned}$$

Therefore, the generating function $X(z)$ satisfies the following ODE

$$X'(z) - \left(\frac{2}{z} + \frac{az}{1-z} \right) X(z) = \frac{bz^3}{(1-z)^2}.$$

We have $\mu(z) = \exp \left(\int - \left(\frac{2}{z} + \frac{a}{1-z} - a \right) dz \right) = z^{-2}(1-z)^a$. which gives

$$z^{-2}(1-z)^a X(z) = \int b(1-z)^{a-2} dz.$$

There are two cases of a that determine the integral. If $a = 1$, we have

$$z^{-2}(1-z)X(z) = b \ln \left(\frac{1}{1-z} \right) + C.$$

Since $x_1 = x_2 = 0$, we have $z^{-2}X(z) \rightarrow 0$ as $z \rightarrow 0$, which gives $C = 0$. Therefore, we have

$$X(z) = b \frac{z^2}{1-z} \ln \left(\frac{1}{1-z} \right).$$

Hence, $x_n = bH_{n-2} \sim b \ln n$.

If $a \neq 1$, we have

$$z^{-2}(1-z)^a X(z) = -\frac{b}{a-1}(1-z)^{a-1} + C.$$

Using the same argument as above, we have $C = \frac{b}{a-1}$. Therefore, we have

$$X(z) = \frac{b}{a-1} z^2 \left((1-z)^{-a} - (1-z)^{-1} \right).$$

Using the generalized binomial theorem

$$(1-z)^{-a} = \sum_{n=0}^{\infty} \binom{n+a-1}{n} z^n,$$

we have

$$x_n = \frac{b}{a-1} \left(\binom{n-3+a}{n-2} - 1 \right) \sim \frac{b}{a-1} \left(\frac{n^{a-1}}{\Gamma(a)} - 1 \right).$$

If $a < 1$ then $x_n \sim \frac{b}{1-a}$. If $a > 1$, then $x_n \sim \frac{b}{(a-1)\Gamma(a)} n^{a-1}$. □

Next, we want to see if adding a difference term to the recurrence changes the asymptotic behavior. Consider the recurrence

$$x_{n+2} = \frac{a}{n} \sum_{i=1}^n x_i + \frac{b}{n} \sum_{i=1}^n |x_i - x_{n+1-i}| + c, \quad n \geq 1, \quad x_1 = x_2 = 0. \quad (2.2)$$

where $a > 1$, $0 < b < 1$ and $c > 0$ are constants.

Lemma 3. *The sequence (x_n) is increasing.*

Proof. One checks that $x_n \geq 0$ for all n . Let us prove that (x_n) is increasing. We have $x_2 \geq x_1$.

Assume that $x_n \geq x_{n-1} \geq \dots \geq x_0$ for some $n \geq 3$. Then,

$$\begin{aligned}
nx_{n+2} &= a \sum_{i=1}^n x_i + b \sum_{i=1}^n |x_i - x_{n+1-i}| + nc \\
&= a \sum_{i=1}^n x_i + b \sum_{i=1}^{\lfloor n/2 \rfloor} (x_{n+1-i} - x_i) + nc \\
&= a \sum_{i=1}^n x_i + b \sum_{i=1}^{\lfloor n/2 \rfloor} (x_{n+1-i} - x_i) + nc \\
&= a \sum_{i=1}^n x_i + b \sum_{i=1}^{\lfloor n/2 \rfloor} (x_{n+1-i} - x_i) + nc \\
&= a \sum_{i=1}^n x_i + b \sum_{i=1}^{\lfloor n/2 \rfloor} (x_{n+1-i} - x_i) + nc \\
&= (a+b) \sum_{i=1}^n x_i - 2b \sum_{i=1}^{\lfloor n/2 \rfloor} x_i - bm_n + nc,
\end{aligned}$$

where

$$m_n = \begin{cases} x_{\lfloor n/2 \rfloor + 1} & \text{if } n \text{ is odd,} \\ 0 & \text{if } n \text{ is even.} \end{cases}$$

Let $S_n = \sum_{i=1}^n x_i$. Then,

$$\begin{aligned}
nx_{n+2} &= (a+b)S_n - 2bS_{\lfloor n/2 \rfloor} - bm_n + nc, \\
S_{n+1} &= S_n + x_{n+1}.
\end{aligned}$$

Hence,

$$nx_{n+2} - (n-1)x_{n+1} = (a+b)x_n - 2b(S_{\lfloor n/2 \rfloor} - S_{\lfloor (n-1)/2 \rfloor}) - b(m_n - m_{n-1}) + c.$$

Let us check the right-hand side according to the parity of n .

- If $n = 2k$, then $S_{\lfloor n/2 \rfloor} - S_{\lfloor (n-1)/2 \rfloor} = x_k$ and $m_n - m_{n-1} = -x_k$. The right-hand side is $(a+b)x_n - 2bx_k + c$.
- If $n = 2k+1$, then $S_{\lfloor n/2 \rfloor} - S_{\lfloor (n-1)/2 \rfloor} = 0$ and $m_n - m_{n-1} = x_{k+1}$. The right-hand side is $(a+b)x_n - 2bx_{k+1} + c$.

Collecting the two cases, we have

$$nx_{n+2} - (n-1)x_{n+1} = (a+b)x_n - 2bx_{\lfloor n/2 \rfloor} + c.$$

The proof follows from Proposition 4. □

Lemma 4. *The sequence can be rewritten as*

$$nx_{n+2} - (n-1)x_{n+1} = (a+b)x_n - 2bx_{\lfloor n/2 \rfloor} + c. \quad (2.3)$$

Induction

Consider the sequence (x_n) defined by

$$nx_{n+2} - (n-1)x_{n+1} = (a+b)x_n - bx_{\lceil n/2 \rceil} + c, n \geq 1 \quad x_0 = x_1 = x_2 = 0, \quad (3.1)$$

where a, b and c are constants such that $a \geq 1, 0 \leq b, c \leq 1$ and $a+b \leq 2$. Suppose that $x_n \geq 0$, for all $n \geq 3$.

Note that from the recurrence (2.2), we easily have $x_n \geq 0$. If we assume that (x_n) is increasing, then we get (3.1). So we can assume $x_n \geq 0, \forall n \geq 1$ and continue to prove that (x_n) is increasing. We currently cannot prove that $x_n \geq 0$ for all $n \geq 1$ right from (3.1).

3.1 Increasingness

Proposition 4. *Consider the sequence (x_n) defined by*

$$nx_{n+2} - (n-1)x_{n+1} = (a+b)x_n - bx_{\lceil n/2 \rceil} + c, n \geq 1 \quad x_0 = x_1 = x_2 = 0, \quad (3.2)$$

where a, b and c are constants such that $a \geq 1, 0 \leq b, c \leq 1$ and $a+b \leq 2$. Suppose that $x_n \geq 0$, for all $n \geq 3$. Then (x_n) is increasing.

3.1.1 Analysis

Let $\Delta_n = x_{n+1} - x_n$, we have $\Delta_1 = 0$ and $\Delta_2 = c$. Assume that $\Delta_k \geq 0$ for all $k \leq n$. We have to show that $\Delta_{n+1} \geq 0$.

Rewrite 4 as

$$n(x_{n+2} - x_{n+1}) + x_{n+1} - x_n = (a+b-1)x_n - bx_{\lceil n/2 \rceil} + c. \quad (3.3)$$

Let $t_n = (a+b-1)x_n - bx_{\lceil n/2 \rceil} + c$. Then

$$t_{n+1} - t_n = (a+b-1)(x_{n+1} - x_n) - b(x_{\lceil (n+1)/2 \rceil} - x_{\lceil n/2 \rceil}).$$

Therefore, we have a new system

$$\begin{cases} n\Delta_{n+1} + \Delta_n = t_n, n \geq 3 \\ t_{2k} - t_{2k-1} = (a+b-1)\Delta_{2k-1}, k \geq 2 \\ t_{2k+1} - t_{2k} = (a+b-1)\Delta_{2k} - b\Delta_k, k \geq 2 \end{cases}$$

From $\Delta_{n+1} = \frac{t_n - \Delta_n}{n}$, we have an equivalence of two unproved statements

$$\Delta_{n+1} \geq 0 \iff t_n \geq \Delta_n.$$

Also from $\Delta_{n+1} = \frac{t_n - \Delta_n}{n}$ we have the equivalence of two granted statements

$$\Delta_{n-1} \geq 0 \iff \Delta_n \leq \frac{t_{n-1}}{n-1}.$$

If we can prove that $(n-1)t_n \geq t_{n-1}$, we are done. This is true for even n according to the second equation of the system. For $n = 2k+1$, we have

$$\begin{aligned}
2kt_{2k+1} - t_{2k} &= (2k-1)t_{2k+1} + (a+b-1)\Delta_{2k} - b\Delta_k \\
&\geq (2k-1)[(a+b-1)x_{2k+1} - bx_{k+1} + c] - b\Delta_k & (\Delta_{2k} \geq 0) \\
&\geq (2k-1)[(a-1)x_{2k+1} + c] - b\Delta_k & (x_{2k+1} \geq x_{k+1}) \\
&\geq (2k-1)[(a-1)(\Delta_k + c) + c] - b\Delta_k & \left(x_{2k+1} = \sum_{i=1}^{2k} \Delta_i\right) \\
&= (2k-1)ac + [(2k-1)(a-1) - b]\Delta_k
\end{aligned}$$

Note that to use $x_{2k+1} \geq \Delta_k + \Delta_2 = \Delta_k + c$, we need $k \geq 3$. So we have to check the cases $n = 3, 4, 5$ separately. Now we continue.

If $a-1 \geq b$, we are done.

If $a-1 < b$. We will show that $(2k-1)ac \geq b\Delta_k$, or stronger, $\Delta_k \leq (2k-1)c$. That leads us to define a stronger inductive hypothesis

$$0 \leq \Delta_k \leq (2k-1)c, \forall 1 \leq k \leq n.$$

With that hypothesis, the lower bound is proved as before. For the upper bound, we have

$$\begin{aligned}
\Delta_{n+1} &= \frac{t_n - \Delta_n}{n} \leq \frac{t_n}{n} & (\Delta_n \geq 0) \\
&= \frac{(a+b-1)x_n - bx_{\lceil n/2 \rceil} + c}{n} \leq \frac{x_n + c}{n} & (x_{\lceil n/2 \rceil} \geq 0, a+b \leq 2) \\
&= \frac{1}{n} \left(\sum_{i=1}^{n-1} \Delta_i + c \right) \\
&\leq \frac{1}{n} \left(\sum_{i=1}^{n-1} (2i-1)c + c \right) & (\Delta_i \leq (2i-1)c, \forall i \leq n) \\
&= c \frac{(n-1)^2 + 1}{n} \leq (2n+1)c.
\end{aligned}$$

3.1.2 Proof

Proof. Let $\Delta_n = x_{n+1} - x_n$ and $t_n = (a+b-1)x_n - bx_{\lceil n/2 \rceil} + c$. We have

$$\Delta_1 = 0, \Delta_2 = c, t_1 = c, t_2 = (a+b)c, \quad \text{and}$$

$$\begin{cases} n\Delta_{n+1} + \Delta_n = t_n, n \geq 3 \\ t_{2k} - t_{2k-1} = (a+b-1)\Delta_{2k-1}, k \geq 2 \\ t_{2k+1} - t_{2k} = (a+b-1)\Delta_{2k} - b\Delta_k, k \geq 1 \end{cases} \quad (3.4)$$

Let $d = a+b$, then $1 \leq d \leq 2$. We have

$$\begin{aligned}
\Delta_3 &= \frac{1}{2}(d-1)c, & t_3 &= (2d-1)c \\
\Delta_4 &= \frac{1}{6}(3d-1)c, & t_4 &= \left(\frac{1}{2}(d-1)^2 + (2d-1) \right) c \\
\Delta_5 &= \frac{1}{24}(3(d-1)^2 + 9d-5)c.
\end{aligned}$$

Assume that $0 \leq \Delta_m \leq (2m-1)c$ for all $1 \leq m \leq n$. We have to show that $0 \leq \Delta_{n+1} \leq (2n+1)c$.

If $n = 2k$, we have $(2k-1)t_{2k} - t_{2k-1} \geq t_{2k} - t_{2k-1} = (a+b-1)\Delta_{2k-1} \geq 0$

$$\text{Hence } \Delta_{2k} = \frac{t_{2k-1} - \Delta_{2k-1}}{2k-1} \leq \frac{t_{2k-1}}{2k-1} \leq t_{2k}.$$

$$\text{Hence } \Delta_{2k+1} = \frac{t_{2k} - \Delta_{2k}}{2k} \geq 0.$$

If $n = 2k+1$, we have $t_{2k+1} = (a+b-1)x_{2k+1} - bx_{k+1} + c \geq (a-1)x_{2k+1} + c \geq c$.

$$\text{Hence } b\Delta_k \leq (2k-1)c \leq (2k-1)t_{2k+1}$$

$$\text{Hence } 2kt_{2k+1} - t_{2k} = (2k-1)t_{2k+1} + (a+b-1)\Delta_{2k} - b\Delta_k \geq 0.$$

$$\text{Hence } \Delta_{2k+1} = \frac{t_{2k} - \Delta_{2k}}{2k} \leq \frac{t_{2k}}{2k} \leq t_{2k+1}.$$

$$\text{Hence } \Delta_{2k+2} = \frac{t_{2k+1} - \Delta_{2k+1}}{2k+1} \geq 0.$$

The lower bound is proved. For the upper bound, we have to do exactly the same calculation as in the analysis section.

$$\begin{aligned} \Delta_{n+1} &= \frac{t_n - \Delta_n}{n} \leq \frac{t_n}{n} & (\Delta_n \geq 0) \\ &= \frac{(a+b-1)x_n - bx_{\lceil n/2 \rceil} + c}{n} \leq \frac{x_n + c}{n} & (x_{\lceil n/2 \rceil} \geq 0, a+b \leq 2) \\ &= \frac{1}{n} \left(\sum_{i=1}^{n-1} \Delta_i + c \right) \\ &\leq \frac{1}{n} \left(\sum_{i=1}^{n-1} (2i-1)c + c \right) & (\Delta_i \leq (2i-1)c, \forall i \leq n) \\ &= c \frac{(n-1)^2 + 1}{n} \leq (2n+1)c. \end{aligned}$$

□

3.2 Asymptotic

Proposition 5. *The generating function of the sequence (x_n) is given by*

$$z(1-z)X'(z) + (2z-2-(a+b)z^2)X(z) = -bz(1+z)X(z^2) + \frac{cz^3}{1-z} \quad (3.5)$$

3.2.1 Analysis

From (2.2), we can ignore the different terms and arrive at a polynomial growth. So we guess that $x_n \sim Dn^\alpha$ for some constants D and α , that is $x_n = Dn^\alpha + o(n^\alpha)$. However, x_{n+2} and x_{n+1} are multiplied by a linear term, so let us assume a stronger hypothesis $x_n = Dn^\alpha + o(n^{\alpha-1})$. Then,

$$\begin{aligned} nx_{n+2} &= n(D(n+2)^\alpha + o(n^{\alpha-1})) = Dn^{\alpha+1} \left(1 + \frac{2}{n} \right)^\alpha + O(n^\alpha) \\ &= Dn^{\alpha+1} \left(1 + \frac{2\alpha}{n} + o\left(\frac{1}{n^2}\right) \right) + o(n^\alpha) \\ &= Dn^{\alpha+1} + 2\alpha Dn^\alpha + o(n^\alpha). \end{aligned}$$

Similarly,

$$(n-1)x_{n+1} = nx_{n+1} - x_{n+1} = Dn^{\alpha+1} + (\alpha-1)Dn^{\alpha} + o(n^{\alpha}).$$

Hence,

$$nx_{n+2} - (n-1)x_{n+1} = D\alpha n^{\alpha} + o(n^{\alpha}).$$

On the other hand, we have

$$(a+b)x_n - 2bx_{\lceil n/2 \rceil} + c = D \left((a+b)n^{\alpha} - \frac{2b}{2^{\alpha}}n^{\alpha} \right) + o(n^{\alpha}).$$

So we must have

$$\alpha + 1 = a + b - b2^{-\alpha+1}.$$

Experiments support this.

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